

# Section 99 — Master Frontier DP Mixed Contest

## Final contest contract

Six original problems cover the complete frontier-DP block.

| Problem | State boundary                                  |
|---------|-------------------------------------------------|
| A       | tight prefix, digit sum, and remainder          |
| B       | tight prefix, digit sum, and KMP state          |
| C       | periodic obstacle-profile transfer matrix       |
| D       | canonical component labels with forced cells    |
| E       | reusable answers for every terminal subset      |
| F       | vertex-cover complement on a tree decomposition |

The final habit is explicit forgetting: state compression is correct only after proving that discarded information cannot affect any future transition.

## State-boundary workflow

1. Fix a processing order and draw the boundary between processed and unprocessed input.
2. List every way a future choice can interact across that boundary.
3. Store exactly the equivalence class needed to answer those interactions.
4. Define introduce, local update, and forget transitions.
5. Prove that two histories with the same stored state have identical valid futures and identical remaining costs or counts.

The order is part of the algorithm. A narrow grid frontier yields exponential dependence on width, while a poor orientation may exponentiate the height. A tree decomposition replaces geometric width by bag width, but the same rule holds: exponential state belongs only on the active separator.

## Choosing among the six patterns

Digit DP uses a prefix boundary and finite automata. Broken-profile tiling uses occupancy masks because no global connection matters. Plug DP adds canonical component labels when processed pieces must connect later. Steiner DP uses a terminal-subset dimension and shortest-path closure. Nice-tree-decomposition DP uses bag assignments when the separator is supplied explicitly. Transfer matrices apply when the same local boundary transition repeats many times.

## Final stress discipline

Build exhaustive oracles before optimizing: enumerate bounded integers, all small tilings, cell subsets, terminal-connecting edge subsets, or vertex subsets. Include empty boundaries, all-blocked rows, components disappearing too early, terminal masks of size zero and one, introduce/forget nodes at the root, and repeated periods whose exponent is zero or one.

## Exercises

A–F form the complete mixed contest.